tf.compat.v1.linalg.tensor_diag_part Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/linalg/tensor_diag_part

> *Returns the diagonal part of the tensor.*

Function Signatures

```
tf.compat.v1.linalg.tensor_diag_part( input, name=None )
```

Code Examples

```
tf.compat.v1.linalg.tensor_diag_part(
input, name=None
)
```


```
tf.compat.v1.linalg.tensor_diag_part(
input, name=None
)
```


```
x = [[[[1111,1112],[1121,1122]],
[[1211,1212],[1221,1222]]],
[[[2111, 2112], [2121, 2122]],
[[2211, 2212], [2221, 2222]]]
]
tf.linalg.tensor_diag_part(x)

tf.linalg.diag_part(x).shape
TensorShape([2, 2, 2])
```

```
x = [[[[1111,1112],[1121,1122]],
```

```
[[1211,1212],[1221,1222]]],
```

```
[[[2111, 2112], [2121, 2122]],
```

```
[[2211, 2212], [2221, 2222]]]
```

Documentation

Returns the diagonal part of the tensor.

This operation returns a tensor with the diagonal part
of the input. The diagonal part is computed as follows:

Assume input has dimensions [D1,..., Dk, D1,..., Dk], then the output is a tensor of rank k with dimensions [D1,..., Dk] where:

diagonal[i1,..., ik] = input[i1, ..., ik, i1,..., ik].

For a rank 2 tensor, linalg.diag_part and linalg.tensor_diag_part produce the same result. For rank 3 and higher, linalg.diag_part extracts the diagonal of each inner-most matrix in the tensor. An example where they differ is given below.

Returns